

Joshua Aryeetey

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EDUCATION

Toronto Metropolitan University (formerly Ryerson University)

Sept. 2024 - Apr. 2028

Bachelor of Engineering, Computer Engineering (Software Specialization)

Toronto, Ontario, Canada

- Dean's List, Merit-Based Academic Scholarship
- Selected Coursework: Algorithms & Data Structures, Software Systems, Database Systems I, Microprocessor Systems

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL

Application & Backend: React Native, Expo, Node.js, Express, FastAPI, REST APIs, Prisma, Streamlit

Data & AI/ML: PostgreSQL, MongoDB, Pinecone, Pandas, NumPy, PyTorch, TensorFlow, RAG, Embeddings, Vector Search

Testing & Tools: Git, GitHub, Docker, Vitest, Jest, pytest, Playwright, Railway, Firebase Auth, Azure, Power BI, Quartus

TECHNICAL EXPERIENCE

AI/ML Research Intern | Generative Molecular Modeling

Mar. 2025 - Jun. 2025

Living in Silico

Toronto, Ontario, Canada

- Built a DeepMol/RDKit molecular-generation workflow over 15,696 rows containing 14,487 unique SMILES, including CSV preprocessing and radius-2 128-bit Morgan fingerprints.
- Trained an RNN molecular generator for 10 epochs with batch size 64 and generated 500 SMILES samples for downstream inspection, comparison, and molecular-generation experiments.
- Explored fragment-based molecular design with RDKit and Fragsmenstein, decomposing structures into reusable substructures and recombining structurally compatible fragments into candidate molecules.

Software Engineering Intern | Data Pipelines & Automation

Sep. 2025 - Nov. 2025

Stush Patties

Toronto, Ontario, Canada

- Built Python ingestion and parsing for Koyo, UNFI, and Dovre distributor reports across CSV, XLSX, and XLSB as part of a two-person client team, converting inconsistent files into a repeatable reporting workflow.
- Normalized SKUs, sales, units, regions, case-pack data, and reporting periods into a canonical schema with source-aware fallback and explicit quality handling, including a pragmatic position/cell parser for the hardest Koyo format.
- Produced unified outputs, data dictionaries, and quality reports that powered repeatable Power BI analysis by distributor, retailer, product, and region.

PROJECTS

Food Tracker - Full-Stack Nutrition & AI Platform | React Native, TypeScript, Express, PostgreSQL, Pinecone, Docker

- Built a full-stack nutrition platform with React Native/Expo and TypeScript/Express APIs backed by PostgreSQL/Prisma, supporting food and weight logging, search, barcode capture, analytics, recommendations, and AI-assisted text/photo input.
- Designed benchmark-driven retrieval across 12,363 reference foods using PostgreSQL fuzzy search and Pinecone candidate generation with deterministic ranking, authoritative rehydration, and versioned index activation/rollback.
- Validated reliability across 1,181 API tests and 584 mobile tests, then hardened staging and deployment with Firebase Auth, Railway health/readiness checks, and production-oriented failure handling.

Cho'Veigo - AI Job Discovery & Resume Tailoring Platform | Python, FastAPI, Streamlit, Gemini

- Co-developed an evidence-based job discovery and resume-tailoring platform with Greenhouse/Lever ingestion, deterministic fit evaluation, Gemini structured outputs, persisted feeds, and automated DOCX/PDF generation.
- Redesigned scoring around responsibility-level evidence and built a human-reviewed benchmark spanning calibration, validation, and sealed cases, reaching 95% exact fit-grade and 100% eligibility agreement on documented validation.
- Hardened first-party job retrieval with static-first requests, bounded Playwright fallback, retries/timeouts, source-health tracking, and deterministic regression coverage; the affected Jobs suite recorded 430 passing tests.

Crest - AI Expense & Fraud Intelligence Platform (3rd Place, MPC Hacks) | Python, FastAPI, MongoDB, Gemini

- Built backend workflows on a four-person hackathon team, processing 4,235 anonymized expense transactions with FastAPI/MongoDB and deterministic policy and preapproval logic.
- Implemented anomaly-detection rules and RAG policy retrieval by chunking policy PDFs, embedding with Gemini, storing 3,072-dimensional vectors in MongoDB Atlas, and retrieving the top 5 chunks from 100 candidates; presented the system to judges, helping earn 3rd place in the Brim Financial Challenge.